

Review

Radio Frequency Interference, Its Mitigation and Its Implications for the Civil Aviation Industry

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Abstract: Radio Frequency Interference has emerged as a growing challenge for aviation safety and system integrity due to the increasing spectral overlap between communication technologies and aviation systems. This paper investigates the sources, types, and consequences of RFI in Global Navigation Satellite Systems, Instrument Landing Systems, and altimeters used in civil aviation. A detailed examination of both intentional and unintentional interference is presented, highlighting real-world incidents and simulated impact models. The study analyzes technical mechanisms such as receiver desensitization, intermodulation, and cross-modulation, and further explores UAV-based interference detection frameworks. Mitigation strategies are reviewed, including regulatory practices, spectrum filters, shielding architectures, and dynamic UAV sensing systems. Comparative insights into simulation results, shielding techniques, and regulatory gaps are discussed. The paper concludes with recommendations for enhancing current aviation standards and suggests a hybrid validation model combining in-flight measurements with simulation-based assessments. This research contributes to the understanding of electromagnetic vulnerabilities in aviation and provides a basis for future mitigation protocols.

Keywords: radio frequency; UAV; VHF; UHF; navigation; aviation; flights; ILS



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1. Introduction

In an era defined by the rapid evolution of wireless communication technologies, the aviation industry faces increasing challenges from RFI. As modern Aircraft heavily rely on electronic and wireless systems for navigation, communication, and surveillance, the integrity of the radio frequency spectrum becomes critical for operational safety and efficiency. However, RFI, whether intentional or unintentional, poses a significant risk to these systems, potentially disrupting flight operations and compromising passenger safety.

The Very High Frequency (VHF) band (118–137 MHz) and Ultra High Frequency (UHF) band (300 MHz–3 GHz) are the backbone of civil aviation communication and navigation systems. VHF is primarily utilized for air-to-ground communication, enabling seamless coordination between pilots and air traffic controllers. Meanwhile, UHF supports a variety of essential functions, including radar systems, transponders, and Instrument Landing Systems (ILSs). Despite strict management regulations, the growing proliferation of wireless technologies, such as 5G networks, Unmanned Aerial Vehicles (UAVs), and Internet of Things (IoT) devices, has increased the likelihood of interference within these critical bands.

RFI can arise from a multitude of sources. External sources include terrestrial communication towers, 5G base stations, improperly shielded electronic equipment, and atmospheric

phenomena such as solar flares or sporadic E-layer activity. Internally, onboard electronic devices, including malfunctioning avionics or improperly used passenger devices, can also contribute to interference. Natural sources, such as ionospheric activity or solar flares, can further degrade or distort signals in the VHF and UHF bands. Regardless of the source, the impacts of RFI on aviation systems are profound, leading to communication failures, navigation errors, and surveillance system malfunctions. For instance, interference with radar altimeters or GPS signals can disrupt flight paths, particularly during critical phases such as landing or take-off, where precision is paramount.

Additionally, intentional electromagnetic interference (IEMI) carried out by malicious actors poses a growing threat to civil aviation. Such threats often involve directed high-power RF signals or electromagnetic pulse devices that can disrupt avionics. Interference may enter systems via front-door coupling (e.g., antennas) or back-door coupling (e.g., power lines, cabling, or structural seams), each requiring different mitigation strategies. According to Jie et al. (2024) [1], the immunity of Aircraft systems must span a wide spectral range—from MHz to several GHz—to ensure resilience across all potential interference scenarios. These requirements are consistent with aviation standards such as RTCA DO-160 [2] and MIL-STD-461 [3], which provide test protocols for radiated and conducted susceptibility in airborne equipment.

Mitigating RFI has therefore become a key area of focus within the aviation industry. Effective mitigation strategies range from advanced technological solutions to stringent regulatory measures. Filters and shielding techniques are integral to protecting onboard systems from interference, while spectrum management ensures that frequency bands allocated to aviation are protected from unauthorized or overlapping transmissions. Additionally, real-time monitoring and enforcement by aviation authorities play a crucial role in identifying and addressing interference sources promptly.

Despite these efforts, significant challenges persist. The global adoption of 5G technology, for example, has brought renewed concerns about its potential to interfere with radar altimeters operating in adjacent frequency bands. Moreover, the increasing use of UAVs and IoT devices introduces complexities in managing the radio frequency spectrum, particularly in urban environments where RF activity is dense. The challenge is further compounded by the limited availability of alternative navigation aids, as many aviation systems rely predominantly on satellite-based technologies like GPS.

Considering these developments, the need for comprehensive research into RFI, its mitigation, and its implications for the aviation industry is more pressing than ever. Understanding the mechanisms of interference, exploring innovative mitigation techniques, and developing robust regulatory frameworks are essential steps in safeguarding the future of aviation. This research aims to contribute to these efforts by investigating the sources, impacts, and mitigation strategies for RFI, with the ultimate goal of enhancing safety and efficiency in the aviation sector.

By addressing these challenges, the aviation industry can not only maintain operational integrity but also adapt to the rapidly evolving technological landscape. This paper will explore these aspects in detail, shedding light on the critical role of RFI management in ensuring the continued safety and reliability of global air travel.

The reliance on advanced electronic and wireless systems in aviation has made RFI a critical challenge. RFI disrupts essential communication, navigation, and surveillance systems, posing risks to safety and efficiency. With emerging technologies like 5G, IoT, and UAVs increasing the likelihood of interference in VHF and UHF bands, addressing RFI has become vital. This research focuses on identifying RFI sources, analyzing its impacts, and proposing mitigation strategies using real-time monitoring and simulations to ensure

aviation system safety and reliability. The primary research objectives of this study are as follows:

- Review multiple research papers covering RFI interference to figure out the methodologies and technologies used in identifying the RFI.
- Identify the sources of RFI in aviation, including external factors like terrestrial transmitters, on-board electronic equipment, portable electronic devices, and weather.

This paper is organized as follows: Section 2 provides the theoretical background, Section 3 provides a literature review, Section 4 provides a discussion of the results, and Section 5 concludes the paper.

2. Theoretical Background

RFI is a phenomenon where unwanted electromagnetic signals disrupt the operation of electronic systems. In aviation, the consequences of such interference are particularly critical, as industry relies heavily on precise and reliable communication, navigation, and surveillance systems. Understanding the theoretical underpinnings of RFI is essential for developing effective mitigation strategies to ensure aviation safety and efficiency.

RF interference in aviation systems may result from several technical mechanisms. One such mechanism is receiver desensitization, where strong out-of-band signals suppress the sensitivity of adjacent weak signals, such as those from navigation beacons. Intermodulation occurs when two or more signals mix in a non-linear component (e.g., an amplifier), generating spurious frequencies that fall within the receiver band. Additionally, cross-modulation may arise when an unwanted signal modulates the amplitude of a desired carrier, leading to distorted reception. These effects are particularly relevant in densely packed RF environments around airports, where multiple emitters coexist in proximity.

2.1. The Role of the RF Spectrum in Aviation

The radio frequency spectrum is the foundation for aviation communication and navigation. The VHF band supports air-to-ground voice communication, enabling pilots and air traffic controllers to coordinate operations effectively. The Ultra High Frequency (UHF) band is used for functions such as radar, transponder communication, and (ILS). These systems ensure safe take-off, cruising, and landing, particularly in poor visibility or adverse weather conditions. RFI can couple into Aircraft systems through two primary mechanisms: front-door and back-door coupling. Front-door coupling refers to interference that enters through designed input ports, such as antennas or receiver terminals. This type is often filtered but still susceptible to strong nearby emissions. In contrast, back-door coupling occurs through unintended paths, including power cables, structural seams, or signal lines, where electromagnetic waves bypass front-end protection. Both coupling types can disrupt onboard avionics, especially if shielding or grounding measures are inadequate. RFI impacts aviation systems through several pathways:

- Adjacent Channel Interference: Signals from neighboring frequency bands bleed into aviation channels, reducing signal clarity.
- Harmonic Interference: Harmonics from non-aviation frequencies can coincide with aviation bands, creating noise and disrupting system performance.
- Receiver Desensitization: Strong external signals can overwhelm receivers, making them less sensitive to desired signals.
- Multipath Propagation: Reflected signals from terrain, buildings, or other surfaces can create signal distortion and loss.

2.2. Implications for Aviation Systems

The impacts of RFI on aviation systems are broad and significant:

- **Communication Systems:** Interference can result in noise, signal loss, or communication delays between pilots and air traffic controllers.
- **Navigation Systems:** Disruptions to GPS or radar altimeters can compromise flight path accuracy, particularly during critical operations such as landing or approach.
- **Surveillance Systems:** Interference with radar systems or transponders can obscure Aircraft positioning, increasing collision risks.
- Recent incidents of radio RFI have significantly impacted aviation operations, highlighting the critical need for robust navigation systems and international cooperation. Two notable examples are given below.

1. **GPS Interference Affecting Finnish Airports (November 2024) [4]:** In November 2024, three airports in Eastern Finland—Joensuu, Savonlinna, and Lappeenranta—experienced disruptions in GPS signals, suspected to be caused by Russian interference. This interference compromised the accuracy of navigation systems, prompting Finavia, the Finnish airport operator, to reinstate traditional radio navigation equipment to assist with Aircraft landings. This situation underscores the vulnerability of satellite-based navigation systems to external disruptions and the necessity for alternative navigation aids to ensure aviation safety.

2. **Unauthorized Radio Transmissions in Melbourne, Australia (September 2024):** Between July and September 2024, a 45-year-old man in Melbourne allegedly made unauthorized radio transmissions, including fake mayday calls, to commercial Aircraft operating at Melbourne Airport. These transmissions had the potential to endanger Aircraft operations by causing confusion and diverting attention from genuine emergencies. The Australian Federal Police arrested the individual, charging him with offenses that carry significant penalties, emphasizing the serious nature of such interference.

The above-mentioned incidents illustrate the diverse sources of RFI—from suspected state-sponsored GPS jamming to individual malicious acts—and their profound impact on aviation safety. They highlight the importance of maintaining and updating traditional navigation aids, enforcing stringent regulations against unauthorized transmissions, and fostering international collaboration to mitigate and respond to RFI threats effectively.

To better understand the safety implications of RFI, fault modes can be classified based on severity and likelihood. For instance, interference with radar altimeters during landing presents a high-severity, moderate-likelihood event due to its impact on altitude accuracy. GPS jamming during enroute flight is moderate in severity but higher in likelihood, especially near conflict zones. Unauthorized radio transmissions fall under moderate severity and low likelihood but can still create confusion. A structured risk matrix can help prioritize mitigation, with high-risk scenarios requiring immediate attention through shielding, redundancy, or alternative aids.

Aviation communication and navigation systems are vulnerable to both unintentional noise and intentional interference. Common noise sources include atmospheric discharge, cosmic background radiation, and electronic emissions from nearby equipment. These noise elements can degrade the signal-to-noise ratio (SNR), leading to reduced positioning accuracy or missed alerts. To mitigate these issues, systems often employ filtering, low-noise amplifiers (LNAs), and error correction algorithms. For safety-critical operations like ILS or GNSS-based approaches, spectrum management and guard band allocation are also essential in reducing noise overlap with active aviation frequencies.

2.3. Mitigation Strategies

Addressing RFI requires a combination of technological, regulatory, and operational approaches: **Technological Solutions:** Filters, shielding, and adaptive antenna designs help mitigate RFI at the system level. Advanced algorithms are also being developed to detect and counteract interference dynamically. **Spectrum Management:** Regulatory bodies such

as EUROCONTROL in Europe enforce strict frequency allocation and emission standards to protect aviation bands from unauthorized use or overlap. Monitoring and Detection: Continuous monitoring of the RF spectrum using tools like spectrum analyzers and UAVs aids in identifying and addressing interference sources.

Recent work also demonstrates the effectiveness of structured safety approaches. For example, Zhang et al. (2021) [5] proposed a three-stage enhancement framework for UAV systems involving electromagnetic shielding, digital filtering, and real-time diagnostics. Similar architectures may be adapted for aviation platforms where mission-critical electronics must withstand burst noise and high-power emissions. Additionally, frequency-hopping spread spectrum (FHSS) has emerged as a resilient technique for reducing susceptibility to narrowband interference, especially in airborne communication links exposed to hostile RF conditions.

2.4. Challenges in Mitigating RFI

Despite advancements in mitigation techniques, several challenges persist [6]. The rapid expansion of wireless technologies, including 5G networks and Internet of Things (IoT) devices, has increased RF spectrum congestion, raising the likelihood of interference. Additionally, the reliance on satellite-based navigation systems like GPS makes aviation systems vulnerable to jamming and spoofing. Developing robust alternatives and enhancing existing systems is critical to overcoming these challenges.

Test Equipment Used to Investigate RFI in Aviation. The detection and mitigation of RFI in aviation require the use of specialized test equipment. These tools are integral for identifying sources of interference, analyzing their impact on aviation systems, and validating mitigation strategies to ensure the safe and efficient operation of Aircraft.

The detection and mitigation of RFI in aviation rely on specialized tools that ensure system integrity and operational safety. These tools help identify interference sources, analyze signal disruptions, and validate mitigation strategies. Their role is crucial in maintaining reliable communication, navigation, and surveillance systems. Details of these tools are discussed below.

2.4.1. Test Equipment Used to Investigate RFI

A spectrum analyzer [7] is a critical tool for monitoring radio frequencies. It measures the power of electromagnetic signals over a range of frequencies, making it possible to detect unauthorized or unintended transmissions that could disrupt aviation communication or navigation systems. Spectrum analyzers are widely used to identify interference in both VHF and UHF bands. RF signal generators [8] are essential for producing controlled electromagnetic signals to test and calibrate aviation systems. These devices help simulate interference scenarios, allowing engineers to evaluate how avionics, such as radar altimeters and communication systems, respond to specific types of interference.

2.4.2. Network Analyzer

A network analyzer [9] is employed to measure the performance of RF components, including antennas and filters. It evaluates parameters such as signal loss and reflection, ensuring that these components are effectively reducing or rejecting unwanted frequencies that could cause interference.

2.4.3. Directional Antennas

Directional antennas [10] are used to locate sources of interference. By focusing on the directionality of signals, these antennas allow the precise tracking of unauthorized transmissions, which is critical for resolving interference issues in airport environments or urban settings.

2.4.4. RF Site Master

An RF Site Master [11] is a portable device used to test the integrity of cable and antenna systems. It identifies issues such as signal degradation or physical damage that could compromise the system's ability to handle interference effectively. These tools are particularly useful for maintaining ground-based navigation aids like ILS.

2.4.5. Radio Communication Test Sets

Radio test sets, such as the CMA 180, are comprehensive testing tools [12] designed for aviation communication systems. These devices assess the performance of air-to-ground radios and their resilience to interference, ensuring the clarity and reliability of voice and data transmissions.

2.4.6. EMC Chambers

EMC chambers are shielded environments where aviation equipment is tested for compliance with electromagnetic interference (EMI) standards. These chambers simulate controlled interference scenarios to determine how well systems perform in real-world conditions.

To ensure that Aircraft systems meet acceptable electromagnetic performance thresholds, industry standards have been developed. The RTCA DO-160 standard is widely used in commercial aviation to define test procedures for environmental conditions, including electromagnetic emissions and susceptibility. In defense applications, MIL-STD-461 outlines detailed EMI/EMC requirements for airborne equipment. These standards guide design, testing, and certification processes to improve interference resilience across mission profiles.

2.4.7. GPS Jamming and Spoofing Detectors

Specialized tools for detecting GPS interference are used to ensure the accuracy of satellite-based navigation systems. These detectors monitor signal quality and identify instances of jamming or spoofing that could disrupt critical navigation functions.

2.4.8. Time Domain Reflectometer (TDR)

TFRs [13] are used to diagnose faults in RF transmission lines. By analyzing the time delay of reflected signals, these tools help locate disruptions in signal pathways that could be contributing to interference.

2.4.9. UAV-Based RF Monitoring Platforms

UAVs equipped with RF sensors provide a mobile platform for interference detection [14]. These systems are especially useful for monitoring interference in areas where traditional ground-based equipment may not be effective, such as airport perimeters or remote locations. The use of these tools ensures that aviation systems remain resilient to interference, enhancing safety and operational reliability. By enabling the precise detection, comprehensive analysis, and validation of mitigation strategies, this equipment plays a pivotal role in protecting critical aviation infrastructure from the risks posed by RFI.

3. Literature Review

RFI is a very broad yet interesting topic; however, when it comes to its implications for the aviation industry, the sourcing of recent research work becomes quite challenging. A lot of effort was put into finding relevant and recent research papers for consideration in this review of the literature (see the citation references at the end of this document).

Bukhari et al. [15] examine the significant concerns about interference of fifth generation (5G) telecommunications technology with aviation altimeters, particularly due to the overlap of frequency bands. This interference poses serious risks to flight operations

and navigational accuracy, potentially compromising aviation safety. In response to these challenges, regulatory authorities around the world have enacted various measures, including the issuance of Airworthiness Directives (AD) and restrictions on the effective isotropic radiated power (EIRP) of 5G base stations near airports. In addition, exclusion zones have been established to mitigate the potential risks associated with 5G operations in proximity to aviation activities. The paper advocates for a collaborative approach among regulators, industry stakeholders, and researchers to develop compatible technologies and regulatory frameworks. It emphasizes the need for ongoing research to better understand interference dynamics and to create comprehensive guidelines that balance the advancement of 5G technology with the critical requirements of aviation safety. Figure 1 shows the proposed Aircraft's altimeter in boresight of a 5G antenna. The orange ellipse in the diagram represents the zone of maximum interference, where the main beam of the 5G interfering signal overlaps with the aircraft's approach path and altimeter signal. This region signifies the critical spatial area in which radio altimeter receivers are most vulnerable to potential signal corruption or degradation due to high-power emissions from 5G base stations. The intersection of the interfering signal's main beam with the aircraft's glide slope and altitude measurement path highlights the likelihood of interference during final approach, especially in low-visibility or instrument landing conditions.

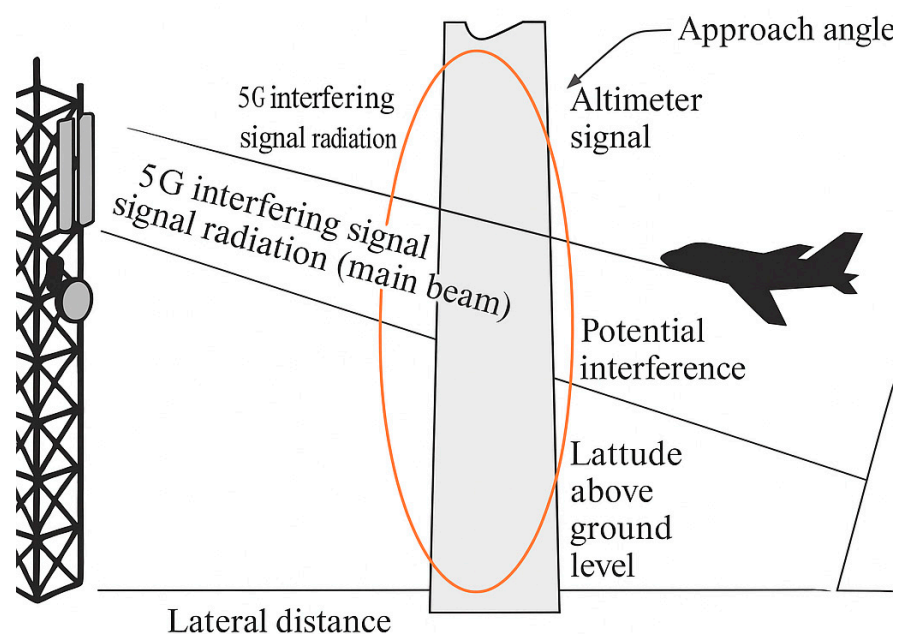


Figure 1. Aircraft's altimeter in boresight of 5G antenna used by Bukhari et al. [14].

Li et al. [16] highlight the substantial interference that 5G base stations can cause to radar altimeters operating within the 4.2–4.4 GHz frequency band, raising significant safety concerns for aviation. A detailed mathematical model is created to systematically assess this interference in various environments, including rural, suburban, and urban areas. The model takes into account different altitudes and operational scenarios to accurately evaluate the interference-to-noise ratio experienced by radar altimeters.

In addition, the study examines the impact of two types of antennas—omnidirectional and multiple input–multiple output (MIMO) antennas—on interference dynamics. This analysis is crucial to understanding how antenna design influences interference. The results emphasize the need for effective mitigation strategies to ensure the safe and reliable coexistence of 5G communication systems and radar altimeters.

Son [17] investigates the potential interference between 5G systems operating within the 3.7–4.0 GHz frequency band and aeronautical radio altimeters operating in the adjacent 4.2–4.4 GHz band in South Korea. This analysis is crucial because South Korea plans to extend the 5G spectrum into the 3.7–4.0 GHz range. The primary objective of the research is to establish technical conditions that ensure compatibility between 5G systems and radio altimeters. This involves evaluating the interference impact of 5G base stations on altimeter operations. The researchers propose an interference analysis scenario and methodology grounded in Monte Carlo simulations. These simulations are designed to replicate real-world conditions, including Aircraft approaches and 5G network deployments. The study identifies the main challenge to determining the necessary restrictions on 5G base station emissions to protect radio altimeter functionality. The findings indicate that unwanted 5G emissions must be 35 dB more stringent than the current 3GPP standards to prevent interference with altimeters. Figure 2 shows the interference scenario in the above-discussed paper, while Figure 3 shows the analysis method.

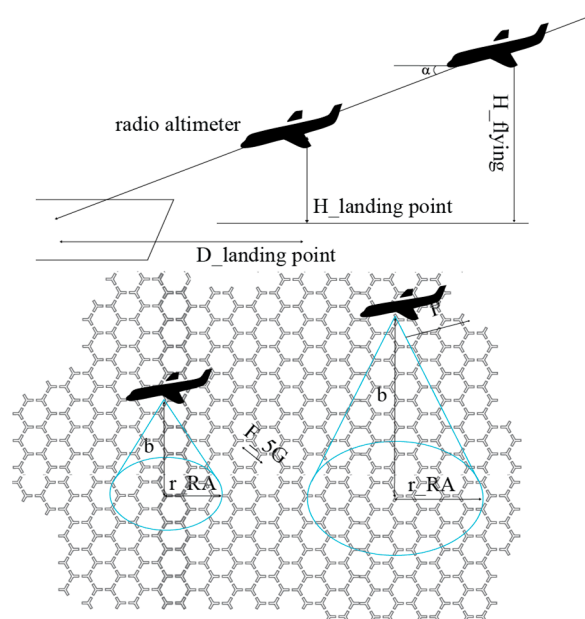


Figure 2. An interference scenario by Son [17]. Schematic representation of the radio altimeter beam spread over a hexagonally divided ground surface, illustrating signal coverage at two different aircraft altitudes. The blue cones indicate the area covered by the altimeter signal, which expands as the altitude increases. In this figure, H represents the altitude of the aircraft above ground level, while D denotes the diameter of the corresponding radio altimeter footprint on the ground. The terms $H_{\text{landing point}}$ and $D_{\text{landing point}}$ refer to the aircraft's altitude and signal footprint, respectively, at the moment of landing.

Zhou et al. [18] conducted a study on identifying sources of civil aviation radio interference using four UAVs. It highlights the limitations of traditional ground-based monitoring methods, which are often slow and ineffective in tracking aerial interference. The study aims to develop an innovative time–frequency difference positioning method to improve monitoring accuracy and efficiency. The research successfully establishes a synchronized model for UAVs and demonstrates its effectiveness through simulations, achieving balanced positioning performance and minimal errors across various deployment configurations. The findings underscore the method's potential for practical applications in aviation safety.

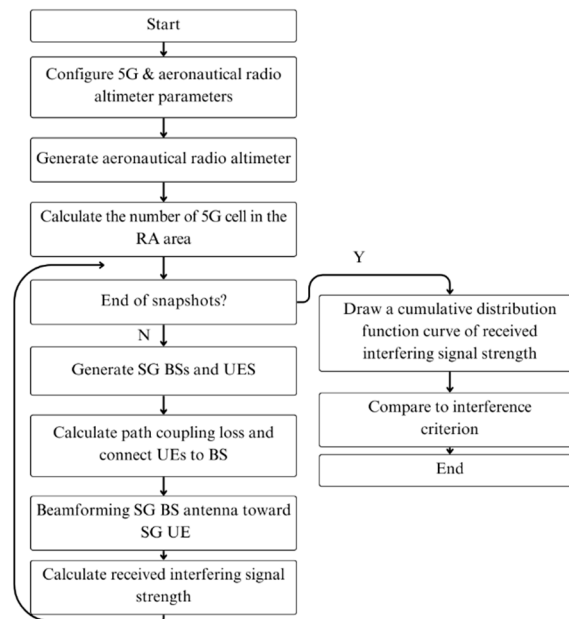


Figure 3. The analysis method used by Son [6].

Kircher, Daniel, and Deutschmann [19] examine how RFI influences the electromagnetic emissions of ICs used in automotive applications. The primary objective is to elucidate how RFI can compromise the functionality and adherence of ICs to EMC standards. The intended readership comprises IC manufacturers and automotive suppliers, particularly those involved in safety-critical applications. A significant challenge addressed is the sequential nature of EMC testing, which frequently overlooks the interplay between electromagnetic immunity and emissions. The results indicate substantial increases in electromagnetic emissions due to injected disturbance signals, resulting in the functional degradation of ICs. Consequently, the findings emphasize the necessity of integrated testing approaches to ensure both safety and regulatory compliance in automotive electronics. Figure 4 provides an example of EMI emissions, while Figure 5 shows the architecture of EMI emissions used by Kircher, Daniel, and Deutschmann [19].

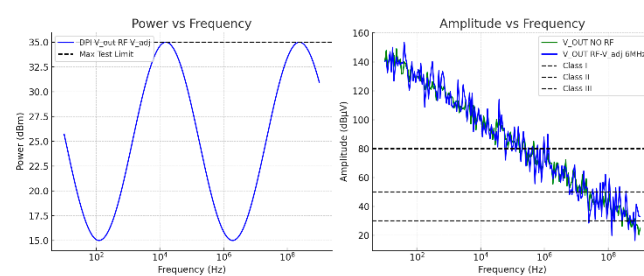


Figure 4. Example of EMI emissions in study by Kircher, Daniel, and Deutschmann [19].

K. Hussain [20] presents a detailed view and explores the critical role of electrical and electronic technology in the twenty-first century, alongside the escalating issue of EMI resulting from the widespread use of electronic devices. It clarifies the often-confused terms EMI and EMC, providing their precise definitions and detailing their interrelationship. The paper underscores that EMI is a disturbance that causes malfunctions or undesired responses in electronic systems, whereas EMC denotes the ability of equipment to operate as intended within its electromagnetic environment. Additionally, it examines the assessment of EMI/EMC and strategies for mitigating EMI issues, with a particular focus on the significance of PCB design in addressing these challenges. For interference mitigation this paper emphasizes implementing bonding, grounding, shielding, and filtering procedures

regularly to address EMI and achieve EMC. It also highlights the significance of best PCB design, the proper construction of structural systems, and addressing communication and radar systems for EMC in naval ships and marine platforms. Furthermore, the paper stresses achieving EMC through design, manufacturing, maintenance, training, and awareness initiatives for improved technological device usage and maintenance.

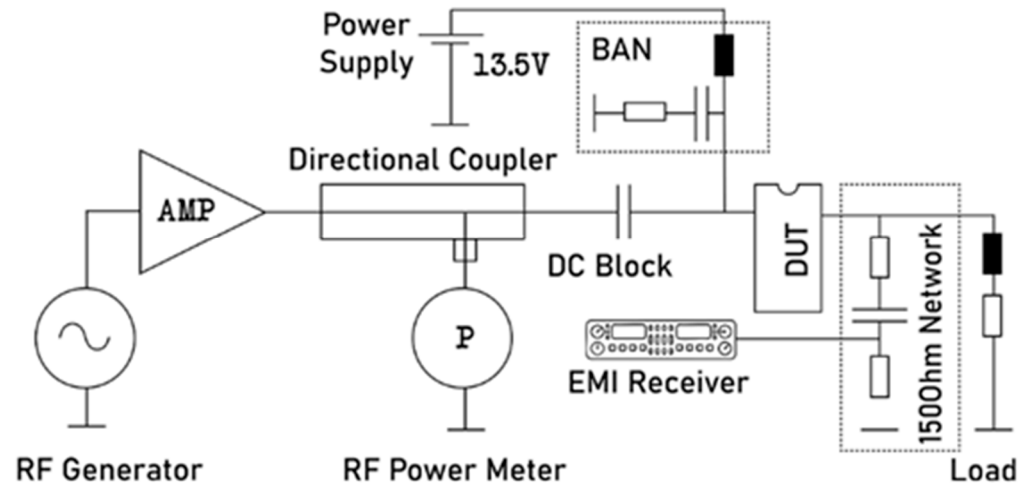


Figure 5. Architecture used by Kircher, Daniel, and Deutschmann [19] to show emissions.

Hegarty et al. [21] examine various mechanisms that cause out-of-band interference, which can disrupt GNSS receivers. It identifies key disruptive factors such as saturation, desensitization, reciprocal mixing effects, intermodulation products, aliasing, and in-band emissions. The paper provides detailed explanations of these mechanisms and discusses strategies to mitigate each one. The study is set against the backdrop of the growing dependence on GNSS technology and the susceptibility of GNSS receivers to out-of-band interference. The findings offer a comprehensive understanding of these disruptive mechanisms and effective mitigation strategies, which are essential for enhancing the reliability and accuracy of GNSSs.

Monitoring interference is an important topic, and Zhou et al. [22] outline a method for monitoring RFI in civil aviation using UAVs, emphasizing four key contributions. Firstly, it introduces an angle-of-arrival (AOA) algorithm that improves the accuracy of RFI location by using directional antennas to determine the azimuth of radio signals. Secondly, it describes the development of a lightweight UAV monitoring platform that meets airworthiness standards, ensuring safe data collection. Thirdly, it details a ground analysis system designed for real-time data processing and the intuitive visualization of RFI locations. Lastly, the method is validated through various test scenarios, showing superior performance in terms of radius monitoring and positioning accuracy compared to traditional ground-based methods. Figure 6 shows the process of Zhou et al. [22] for RFI monitoring using a UAV.

Hosokawa et al. [23] introduce a monitoring network established in Japan to observe the anomalous propagation (EsAP) of aeronautical VHF radio waves, which are influenced by sporadic layers of E (Es). Operational since May 2019, the network comprises six stations equipped with software-defined radios that capture signals within the 98 to 118 MHz frequency range. This system facilitates near-real-time monitoring, providing essential data for aviation navigation and communication. The network has successfully identified multiple instances of EsAP, with notable variations in signal intensity across different stations. This capability enables the distinction between equipment malfunctions and interference caused by Es, thereby enhancing operational safety. Ongoing efforts to

develop algorithms for automatic data analysis aim to further improve the visualization and understanding of Es impacts on VHF communications.

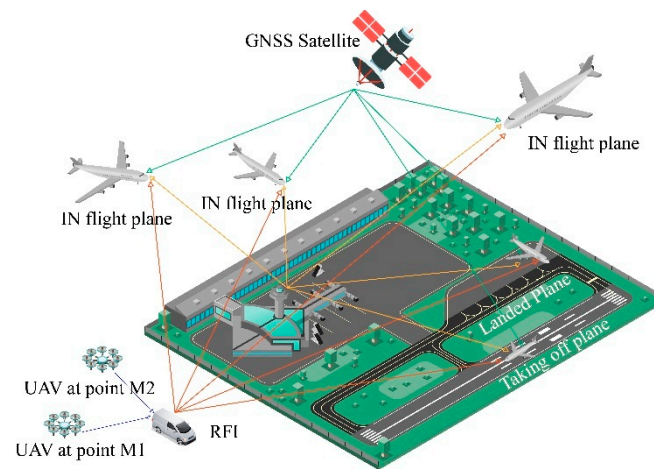


Figure 6. RFI monitoring using UAV in study by Zhou et al. [22].

Joseph et al. [24] explore the growing dangers that RFI poses to Global Navigation Satellite System (GNSS) receivers in commercial Aircraft. They highlight problems such as jamming and spoofing, which can result in navigation failures and safety hazards. To counter these issues, the authors suggest a three-part approach: detecting RFI, accurately measuring the extent of the errors caused, and ensuring that receivers can still provide genuine position information even during interference. Furthermore, the paper stresses the importance of combining GNSS data with other navigation sensors like Inertial Reference Systems (IRSs) and Distance Measuring Equipment (DME) to improve overall navigation integrity and reliability in the face of RFI threats. Figure 7 shows the process of RFI on the GNSS receivers used by the authors.

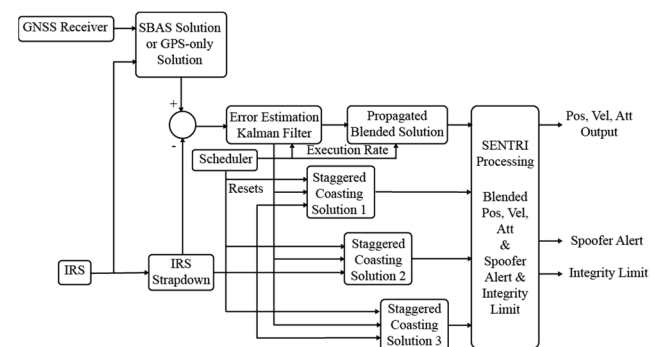


Figure 7. RFI on GNSS receivers used by Joseph et al. [24].

Researchers at MIT [14] discussed three novel techniques to mitigate interference in aeronautical communication: Pulse Blanking, Space Time Block Coding (STBC), and Decision-Directed Noise Estimation (DDNE). Using cognitive radio technology, Aircraft spectrum access was enhanced through dynamic identification and the utilization of available spectral bands, addressing the problem of spectrum scarcity. However, interference from DME presents significant challenges to the LDACS sXignal, as the high power of DME signals can substantially degrade communication quality. The proposed methods aim to effectively reduce this interference, thereby ensuring reliable air-to-ground communication.

Reddy [25] describes an RF front-end combinational filter for airborne V/UHF communication systems which integrates low-pass, high-pass, and band-stop filters, covering a frequency range from 30 to 600 MHz. A specific band-stop filter between 88 and 118 MHz

mitigates interference from commercial FM broadcasts, enhancing communication reliability in software-defined radio (SDR) applications. This design ensures the selection of desired signals while rejecting unwanted interference, preserving signal quality, and improving the overall sensitivity and rejection of interference of the system. Implemented on a Polytetrafluoroethylene (PTFE) Printed Circuit Board (PCB), the filter's compact form factor is suitable for military and defense applications. Comprehensive simulations optimized the filter's performance, and real-world tests closely matched these results, demonstrating effective FM broadcast interference mitigation and high signal integrity. The robust design of the filter improves communication reliability, safety, and operational efficiency in aviation. Figure 8 shows the RF front-end combinational filter used by Reddy [25].

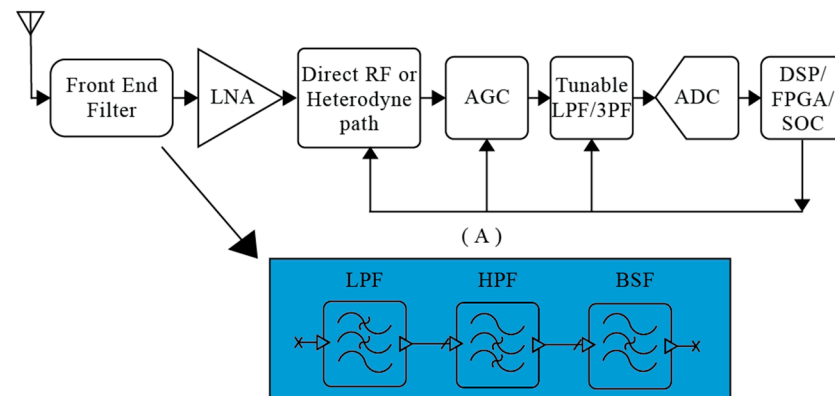


Figure 8. RF front-end combinational filter used by Reddy [25].

Vogel [26] highlights that co-site interference in Aircraft systems arises from several mechanisms, including adjacent channel interference, harmonic frequencies, receiver blocking, intermodulation products, and out-of-band transmission. These issues are particularly prevalent when systems operate across a broad frequency range. To address these challenges, electromagnetic simulation plays a vital role by analyzing the electromagnetic environment and quantifying antenna coupling through S parameters. This study highlights that even modest coordination between systems can significantly improve receiver sensitivity. It underscores the importance of enhancing transmitter electronics and implementing practical mitigation strategies such as narrow bandpass filters and meticulous spectrum management. Ultimately, a thorough understanding of the interference mechanisms, coupled with advanced simulation techniques, leads to effective solutions to optimize communication and sensing systems in Aircraft. Figure 9 shows the co-site interference in an Aircraft system described by Vogel [26]. Table 1 provides the signal strength measurements (db).

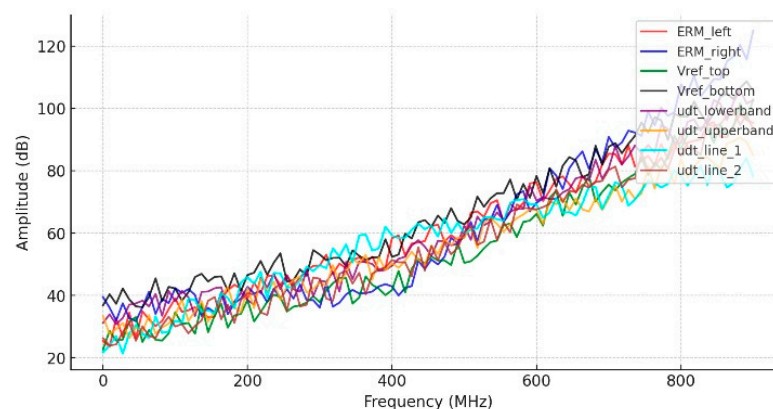


Figure 9. Co-site interference in Aircraft system used by Vogel [26].

Table 1. Signal strength measurements (DB).

Rx Tx	VHF Top	HF	VHF Bottom	UHF Top 1	UHF Top 2	UHF Bottom 1	UHF Lowerband	UHF Bottom 2
VHF top	25.3	−0.8	26.9	28.9	21.8	53.5	45.8	50.1
HF	−0.8	9.8	22.4	22.4	21.8	28.5	28.6	28.6
ESM left	13.4	49.9	28.8	25.8	29.7	38.5	39.5	39.5
ESM right	14.3	43.0	22.0	25.4	31.6	29.3	46.7	46.7
VHF bottom	29.4	27.9	44.3	44.3	52.8	39.5	37.0	27.1
UHF top 1	30.0	39.5	39.8	19.3	19.3	39.8	40.0	40.0
UHF top 2	20.8	39.3	12.3	63.4	−47.4	38.9	39.8	−61.4
UHF bottom 1	39.7	39.9	30.4	63.4	−47.4	40.0	25.6	30.4
UHF lowerband	36.8	39.9	25.7	40.0	40.0	25.6	28.2	30.4
UHF bottom 2	38.9	39.9	25.0	40.0	−30.5	−30.5	27.8	30.4

Ilive [27] describes Intermodulation products (IMPs) arise when multiple signals interact within a non-linear medium, such as electronic components or antennas. This interaction can create new frequencies that are combinations of the original signals, potentially disrupting intended communication or navigation signals. The research highlights that third-order and fifth-order IMPs are particularly troublesome for navigation systems like the ILS and VHF Omnidirectional Range (VOR). These IMPs can produce frequencies that are very close to the primary frequencies used by these systems, thereby increasing the risk of interference. To mitigate these issues, the study emphasizes the need for robust measurement and assessment techniques to identify and evaluate potential conflicts between different communication systems in aviation. This approach aims to ensure that navigational signals remain reliable and safe for Aircraft operations, especially during critical phases such as landing and approach.

Mahama et al. [28] stipulate findings that UAVs utilize specific radio frequency bands, such as 72–73 MHz, 902–928 MHz, and 2400–2483.5 MHz, which are also employed by technologies like Wi-Fi and ZigBee. This overlap can cause radio frequency (RF) interference, particularly in urban settings with numerous devices, increasing the risk of signal disruption. During bridge inspections, this interference can weaken signal strength, degrade video quality, and impair GPS functionality, thus compromising UAV control and safety. To address these challenges, it is essential to choose UAVs with robust RF immunity capable of effectively managing interference. Furthermore, UAVs should have minimal switching delays between redundant communication channels to ensure seamless operation. Conducting a comprehensive assessment of the inspection environment to identify potential RF noise sources is also crucial. By adopting these measures, the reliability and safety of UAV operations during bridge inspections can be improved, leading to successful outcomes even in demanding environments. Figure 10 shows the utilization of a specific frequency band by a UAV.

Kim, Jaesin, and Inkyu Lee [29] examine path loss in long-range air-to-ground (AG) communication systems using the UHF band. They find that the measured path loss aligns well with the predictions of the established models. The UHF band has been shown to be more reliable than higher frequencies, particularly in reducing multipath fading and atmospheric effects, which are crucial for long-distance communication. The analysis reveals that path loss behavior varies with different ground station environments, such as sea and ground reflections, and that multipath components can occur even at long ranges. Flight tests were conducted to observe path loss and multipath characteristics

in realistic scenarios. The setup included UHF antennas, a high-power amplifier, and a channel sounding configuration, allowing data collection over distances of hundreds of kilometers. These findings suggest that UHF communications can be effectively used in UAV systems for military and civilian applications, especially in challenging environments. This research highlights the importance of empirical measurements in developing accurate models for long-range AG communications, aiding in the design and optimization of AG communication systems for reliable UAV connectivity. Figure 11 shows the path loss in long-range A/G communication system.



Figure 10. Utilization of specific frequency band by UAV in study by Mahama et al. [28].

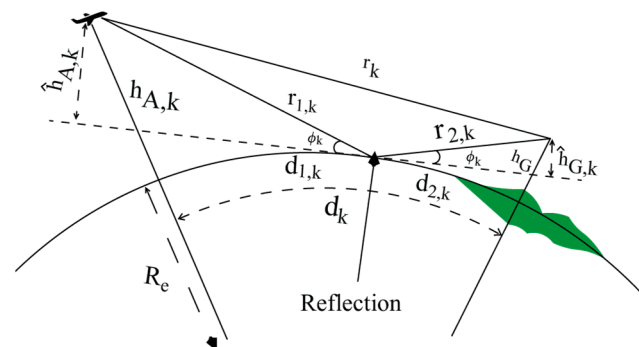


Figure 11. Path loss in long-range A/G communication system in study by Kim, Jaesin, and Inkyu Lee [29].

Xiaoyi [30] describes an ILS, crucial for civil aviation, helping pilots land safely in poor visibility and adverse weather. It uses wireless signals and instruments for precise guidance. Traditionally, maintenance relied on time domain analysis, which can be inaccurate, especially for weak signals. The study highlights the advantages of frequency domain analysis, which accurately measures signal strength across frequencies. Using a spectrum analyzer, maintenance personnel can evaluate the spectral distribution of signals, identifying frequency, power, harmonic components, and noise. This method enhances ILS debugging and calibration, particularly for single-antenna performance and transmitter output adjustments. It also stresses the importance of training to avoid electromagnetic interference and adhering to regulations to reduce external disruptions.

Chunqing [31] highlights the distribution of near-field (NF) signals in localizer antenna systems within a radius of 150 m. It contrasts the NF behavior with far-field (FF) characteristics, underscoring the complexity and intricacy of NF measurements. The study constructs a three-dimensional model, refined through phase analysis and on-site data, to evaluate signal distributions against theoretical predictions. The results indicate that the NF distributions vary significantly with elevation angles and reflection coefficients, with unequal amplitude signals producing higher relative amplitudes at lower angles. In addition, the research examines the effects of Fresnel loss and directional weakening on signal transmission.

According to Elsayem et al. [32] the primary concern with 5G technology is its potential to interfere with radar altimeters due to overlapping frequency ranges (3.7–3.98 GHz for 5G and 4.2–4.4 GHz for radar altimeters). This overlap could pose safety risks during critical flight phases, such as landing and take-off. The study evaluated compatibility by simulating realistic interference scenarios, taking into account various factors, such as the positioning of 5G base stations and the characteristics of radar altimeters. The findings indicated a higher likelihood of interference at lower altitudes, where precise altitude measurements are crucial. To mitigate these risks, the article recommends reducing the transmission power of 5G base stations in specific areas, especially near airports. It also suggests additional strategies such as antenna down-tilting and creating exclusion zones.

Leslie [33] explores the conflict between US regulatory bodies regarding the deployment of 5G technology and its potential effects on Aircraft safety, specifically focusing on radio altimeters. It begins by outlining the planned expansion into C-band frequencies, which has sparked concerns about possible interference with aviation instruments. The methodology includes an analysis of data from various stakeholders, such as the FAA and FCC, to evaluate the risks posed by 5G transmissions. The discussion highlights the importance of cooperation between regulators and industry players to address safety issues while progressing with 5G implementation. It also points out that other countries have managed to roll out 5G with necessary precautions, in contrast to the regulatory hurdles faced in the US. Ultimately, the article advocates for a more data-driven strategy to resolve future disputes over spectrum allocation.

Ivanov et al. [34] developed a deep learning method for detecting GNSS interference using Aircraft data from GPS and AHRS. The model operates in near real-time, analyzing sensor data to identify RFI. It combines temporal sequence modeling with anomaly detection to detect subtle interference patterns missed by rule-based systems. Achieving 83.5% accuracy, it outperforms traditional techniques. Amairah et al. [35] conducted similar work. However, it heavily depended on large labeled training data, which is often unavailable, limiting scalability and deployment.

Studies such as Cevik et al. [36], Joseph et al. [37], and Kennedy et al. [38] used machine learning algorithms with radio spectrum monitoring to classify and mitigate interference in aviation channels. They applied CNNs and RNNs to distinguish legitimate signals from interference, reducing false positives. A multi-tier framework enhanced spectral efficiency and clarity by adjusting detection thresholds based on conditions. However, its high computational cost limits its suitability for low-latency, resource-constrained aviation systems.

Novella et al. [39] introduced a GNSS-jamming protection method that refines interference mask models to improve the detection of intentional jamming. By using probabilistic modeling and adaptive filtering, the method creates accurate interference masks, enhancing jamming protection around critical navigation zones. Simulated tests showed a major decrease in false jamming alerts while maintaining high sensitivity to real threats. However, optimization of the model's computational complexity is needed for real-time use.

Liu et al. [40] studied UAV interference in urban settings using deep learning detection methods, noting the vulnerability of command links to electromagnetic environments. They relied on predefined models, lacking adaptive real-world testing. Wang et al. [41] investigated cognitive radio for mitigating aviation communication interference dynamically with a real-time adaptive prototype. The system is effective but has a high demand for computational resources, limiting its onboard application. Rhoads [42] introduced AIMDS for FAA flight inspection, integrating real-time spectral monitoring with machine learning to enhance RFI tracking and impact assessment on communication and navigation systems. AIMDS used onboard sensors and ground stations for accurate interference localization and classification. However, it did not fully address the challenge of detecting low-power interference below conventional equipment noise floors, which can go unnoticed until avionics performance degrades.

Table 2 presents a comprehensive analysis of the research papers discussed above, addressing various aspects of RFI in aviation. It outlines the technologies utilized, research focus areas, identified gaps, and the equipment or simulations used in each study. The entries cover a wide range of RFI sources, including 5G's impacts on radar altimeters, GNSS system vulnerabilities, and interference from co-located transmitters and receivers. Key themes include the importance of developing effective mitigation techniques, implementing real-time monitoring systems, and improving EMC standards. Identified gaps, such as a lack of field validation, limited automation for anomaly detection, and inconsistencies in regulatory standards, are emphasized. The testing methods highlighted in the table include spectrum analysis, EMI/EMC simulations, UAV-based monitoring, and advanced filtering techniques.

Table 2. Comparison table of various research papers and gap identification.

Ref.	Key Technology Used	Area of Research	Gaps Identified	Test Equipment/Simulation Used
[14]	5G Interference Analysis	Impact of 5G on aviation altimeters	Insufficient real-world implementation data	Airworthiness Directives, EIRP simulation
[15]	Mathematical Modeling	Compatibility of 5G systems with radar altimeters	Need for in-depth field validation	Monte Carlo Simulations
[16]	Interference Scenario Analysis	Compatibility between 5G and aeronautical systems	Challenges in establishing global standards	Simulation tools for operational scenarios
[17]	Time Difference of Arrival (TDOA)	Locating civil aviation RFI sources	Limited deployment in diverse environments	UAV-based synchronization models
[18]	EMC Standards Evaluation	RFI's impact on integrated circuits	Integrated immunity-emission testing methods	EMC spectrum analyzers
[19]	PCB Design for EMI Mitigation	EMC in electronic systems	Integration of PCB-based EMI mitigation techniques	Simulation tools for EMI shielding analysis
[20]	Out-of-Band Interference Mitigation	Impact of interference on GNSS receivers	Broader scenarios for GNSS testing	GNSS signal simulators
[21]	UAV Platforms for RFI Monitoring	Monitoring RFI in aviation using UAVs	Challenges in UAV airworthiness compliance	Angle of Arrival (AOA) algorithms

Table 2. Cont.

Ref.	Key Technology Used	Area of Research	Gaps Identified	Test Equipment/Simulation Used
[22]	Anomalous VHF Wave Monitoring	Sporadic E propagation's impact on VHF signals	Automation in anomaly detection algorithms	Software-defined radios (SDRs)
[23]	GNSSs and Inertial Navigation	Mitigation of RFI in GNSS receivers	Integration of GNSSs with alternative systems	GNSS-DME hybrid models
[24]	Interference Cancellation Techniques	Cognitive radio systems in aeronautical communication	Effectiveness of new algorithms in real-world scenarios	Pulse Blanking, DDNE Simulations
[25]	RF Combinational Filters	Mitigation of FM interference in V/UHF communication	Limited studies on compact, multi-band filter designs	Polytetrafluoroethylene (PTFE) PCB testing
[26]	Electromagnetic Simulations	Co-site interference in Aircraft communication systems	Lack of robust transmitter improvements	S-parameter-based simulations
[27]	Intermodulation Product Analysis	Impact of intermodulation on navigation signals	Better assessment methods for third/fifth-order IMPs	Frequency domain analysers
[28]	RF Immunity Evaluation	Testing UAVs for RF interference during inspections	Limited multi-channel redundancy assessments	RF Noise Source Analysis Tools
[29]	Channel Measurement Tools	Air-to-ground communication using UHF band	Need for extensive testing in varied environments	Channel Sounders, Path Loss Simulations
[30]	Frequency Domain Testing	Instrument Landing System (ILS) maintenance	Broader training for ILS maintenance personnel	Spectrum Analyzers
[31]	Near-Field Signal Modeling	Localizer antenna system signal behavior	Limited understanding of NF vs. FF signal interactions	3D Phase Analysis Tools
[32]	C-Band Interference Mitigation	5G interference with radar altimeters	Implementation challenges for exclusion zones	Simulations of Base Station Placement
[33]	Regulatory Impact Analysis	5G rollout and aviation safety concerns	Data-driven strategies for spectrum allocation	Policy Analysis Tools
[43]	Structured Shielding and Filtering	Multi-layer EMI defense for UAV gimbal systems	Needs adaptation for larger-scale airborne platforms	Real-world prototype, diagnostic filters

4. Discussion and Future Work

The growing reliance on advanced communication technologies, such as 5G, UAVs, and GNSSs, has introduced significant challenges for aviation safety and system performance. RFI poses risks that require a systematic and collaborative approach to mitigate. One of the key challenges is the lack of real-world data on 5G interference mitigation near airports. While theoretical models exist, comprehensive field trials and real-time monitoring are essential to understand the actual impact of 5G signals on radar altimeters and aviation navigation systems. Establishing exclusion zones and validating their effectiveness through data-driven methodologies can help minimize risks.

4.1. Current Research Challenges

Another significant gap is the limited validation of theoretical models assessing interference-to-noise ratios. Many existing studies rely on simulations rather than real-world implementation, making it necessary to conduct field experiments in diverse geographical and environmental conditions to ensure accurate validation. Additionally, disparate regulatory standards across regions create inconsistencies in managing spectrum usage and interference control. A unified framework through international collaborations and shared data can help in establishing standardized mitigation practices, ensuring global aviation safety.

The effectiveness of UAV-based RFI monitoring systems is also an area that needs further research. These systems have been proposed for interference detection, but their testing in varied terrains, such as urban, rural, and high-altitude environments, remains inadequate. Developing modular UAV platforms equipped with adaptive RF monitoring algorithms can optimize interference detection. Additionally, fragmented immunity and emission testing in integrated circuits (ICs) is a concern, as existing methods fail to evaluate immunity and emissions simultaneously. AI-driven predictive models and dual-channel analyzers could enhance these testing processes to ensure compliance with EMC standards.

Another major challenge is the insufficient automation of VHF anomaly detection. It is also important to note that while many mitigation strategies emphasize front-door coupling—such as filtering and antenna shielding—back-door coupling presents more persistent challenges. These pathways often include unintentional entry points through power lines, cable harnesses, or structural seams. Effective protection requires comprehensive grounding, multi-layer shielding, and EMI-aware layout design, which are more complex to implement and validate. The literature still lacks field-tested frameworks specifically focused on this harder-to-control mechanism.

While Mahama et al. [28] propose a multi-layer shielding model for UAVs, their simulations assume ideal ground reflection models and ignore Aircraft cabin resonance. In contrast, Li et al. demonstrate the real-world validation of interference thresholds but only for static scenarios. Our analysis suggests a hybrid validation approach that combines field testing with simulation to achieve scalable, reliable EMI certification. Future standards should mandate real-time in-flight data logging for evaluating signal integrity under operational loads.

Traditional methods rely on manual monitoring, which can be inefficient and error-prone. Deploying machine learning models trained on historical data can help classify and predict anomalies in VHF aviation communications. Similarly, GNSS interference mitigation remains a challenge, as GNSS signals are highly vulnerable to jamming, spoofing, and out-of-band interference. Integrating inertial navigation systems (INS) and advanced signal processing algorithms can enhance resilience against such disruptions.

Managing co-site interference in Aircraft communication systems is another ongoing issue. Existing transmitter and antenna designs are not optimized to minimize interference. Implementing electromagnetic simulations to refine antenna placement and spectrum usage, along with employing adaptive filtering systems, can help mitigate overlapping-frequency problems. Likewise, understanding near-field interactions in localizer antenna systems requires further research, as current studies do not sufficiently differentiate between near-field and far-field behaviors. Conducting real-world experiments alongside simulations would improve knowledge in this domain.

Another gap exists in multi-band RF filtering, as current designs lack versatility. Advanced materials such as Polytetrafluoroethylene (PTFE) could improve filter efficiency. Testing these filters in both laboratory and operational settings will ensure their effectiveness. Furthermore, creating 5G exclusion zones near airports presents logistical challenges.

Using predictive tools to define interference zones dynamically and collaborating with telecom providers to optimize base station configurations could provide a viable solution.

The limited testing of GNSSs under various conditions also poses risks to aviation reliability. Conducting controlled experiments in high-interference environments, such as those caused by multipath effects, would allow for better GNSS validation. Similarly, UAV-based RFI monitoring systems lack certification for airworthiness in aviation. Developing lightweight UAVs with certified airframes for regulatory compliance is essential before they can be fully deployed in aviation applications.

Further research is required in understanding UHF path loss in long-range air-to-ground communication systems, as current models lack sufficient empirical data. Field experiments that assess signal behavior over various environmental conditions can improve aviation communication reliability. Moreover, cognitive radio systems, which dynamically adjust frequency usage, require operational testing to ensure their effectiveness in managing real-time interference.

Another challenge is insufficient training for ILS maintenance personnel in handling advanced signal analysis techniques. Implementing standardized training programs focused on spectrum analyzer usage and interference mitigation strategies could improve maintenance efficiency. Additionally, higher-order intermodulation effects on navigation systems need more in-depth analysis, as interference from intermodulation products could disrupt critical signals like ILS and VOR. Employing frequency domain analyzers for in-depth assessment would help in mitigating such risks.

Testing UAV RF immunity in high-interference environments is another underdeveloped area, as UAVs must be designed to withstand radio frequency noise while performing sensitive aviation-related tasks. Introducing multi-channel redundancy can improve UAV resilience in these environments. Additionally, real-time 5G interference mitigation during flight requires further exploration. Onboard interference filtering mechanisms could help dynamically adjust altimeter readings in response to signal disruptions.

The spectrum allocation strategies in aviation often lack data-driven insights. Establishing a centralized repository for RFI incidents and applying predictive analytics to spectrum allocation policies can ensure more efficient frequency management. By addressing these gaps with advanced research, improved regulations, and technological innovations, aviation systems can become more resilient to RFI threats, ensuring safe and reliable global air travel.

4.2. Future Work

- The field validation of 5G interference exclusion zones using real-time altimeter feedback.
- The certification and deployment of airworthy UAV-based RFI detection platforms.
- Standardized VHF anomaly detection using AI-based classifiers.
- The development of adaptive back-door coupling suppression systems.
- Broader testing of GNSSs in multipath-heavy conditions.
- Lightweight RF immunity designs for UAV and urban inspection tasks.

5. Conclusions

The rising prevalence of RFI in aviation underscores the pressing need for comprehensive strategies to address its impact on safety and operational efficiency. The integration of modern technologies such as 5G, GNSS, and UAV systems has introduced new challenges that demand innovative solutions, rigorous testing, and international collaboration. This study has identified some critical gaps in cited research papers and outlined practical approaches to bridge them.

The issue of 5G interference with radar altimeters highlights the importance of real-world testing and extensive field trials. By leveraging simulation models alongside empirical data, the aviation sector can develop more effective mitigation techniques. Similarly, GNSSs and UAV systems require robust evaluation under diverse environmental and operational conditions to improve their reliability and resilience to interference.

The lack of unified international regulations creates inconsistencies in addressing RFI across different regions. Establishing collaborative platforms for data sharing and policy development can harmonize global standards, and foster consistency and interoperability. Furthermore, integrating artificial intelligence and machine learning into areas such as anomaly detection, spectrum management, and testing protocols offers significant potential to improve efficiency and accuracy.

Technological advancements in materials and simulation tools can help optimize co-site interference management, intermodulation analysis, and multi-band filtering. Compact, high-performance designs and innovative testing methods can enhance system compatibility without compromising efficiency. Additionally, targeted training programs for maintenance personnel and the development of real-time in-flight mitigation tools can strengthen operational capabilities.

Collaboration among industry stakeholders—including regulators, manufacturers, and researchers—is essential for implementing these strategies. By pooling resources and expertise, the aviation industry can establish a unified approach to managing RFI, ensuring technological progress aligns with the need for safety and reliability. Addressing the challenges of RFI requires a proactive, multidisciplinary strategy that incorporates advanced research, innovative technologies, and collaborative policymaking. By following the recommended approaches, the aviation industry can effectively mitigate RFI risks, enhance system robustness, and ensure a safer and more efficient future for global air travel.

Author Contributions: Conceptualization, A.M. and M.R.; methodology, A.M.; software, A.M.; validation, A.M. and M.R.; formal analysis, A.M.; investigation, A.M.; resources, A.M.; data curation, A.M.; writing—original draft preparation, A.M.; writing—review and editing, A.M. and M.R.; visualization, A.M.; supervision, M.R.; project administration, M.R.; funding acquisition, M.R. All authors have read and agreed to the published version of the manuscript.

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